Mapping for Autonomous Driving: Opportunities and Challenges

***Abstract*—This article provides a review of the production and uses of maps for autonomous driving and a synthesis of the opportunities and challenges. For many years, maps have helped human drivers make better decisions, and in the future, maps will continue to play a critical role in enabling safe and successful autonomous driving. There are, however, many technical, societal, economic, and political challenges to mapping that remain unresolved. While fully autonomous driving may be some distance in the future, intermediate steps to realize the technology can be taken. These include developing an efficient and reliable storage and dissemination infrastructure, defining minimum data quality requirements, and establishing an international mapping standard. The article closes with 11 open research challenges for mapping for autonomous driving.**

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For many years, maps have helped human drivers make better decisions during the control and operation of a motor vehicle. These maps, both physical and digital, allow the driver to understand the relationship between their own vehicle and the surrounding environment, in addition to assisting in navigational and routing tasks. With the advancements in autonomous vehicles, maps play an even more crucial role—unlike humans, where simple abstract maps are complementary to the driver's own

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senses, experiences, and judgment, autonomous vehicles require far more detailed maps to aid their decision-making process. Maps can provide an important trusted baseline where the availability of sensors cannot be guaranteed. While opinions differ on the future state of autonomous vehicles, there is consensus that it will challenge transportation norms, infrastructure, and urban development—and that maps will play a critical role in enabling the technology.

Maps can do things that other sensors cannot [57]. First, they have an "infinite range" and, therefore, can "see," even into occluded areas. Second, maps will never fail due to environmental conditions. Within the safety-critical application of autonomous driving, maps can be considered an additional sensor that cannot fail and provide a reliable source of redundancy. Lastly, maps contain highly refined data, which can involve many hours or days of preprocessing and human verification to reduce noise and uncertainty. In turn, this allows maps to provide accurate, meaningful, and useful information in real time.

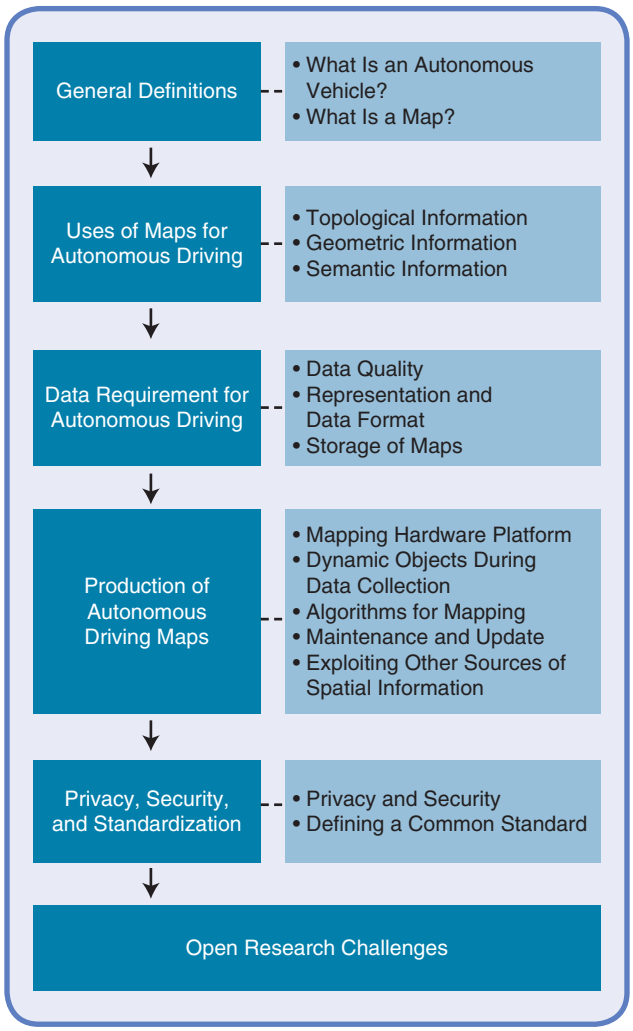

FIG 1 A flowchart of the structure of the article.

Despite these advantages, maps are not infallible. Maps can be inaccurate, inconsistent, or simply out of date. With the higher accuracy requirements of autonomous driving, mapping-related data files can also be very large, introducing challenges in the transmission of updates in real time. Furthermore, the production of mapping of such quality can also incur a high cost.

This article provides a review of the opportunities and challenges of mapping for autonomous driving. A state-of-the-art review of the literature from the past five years was conducted and provides the principal source of surveys. Some industry reports and sources were also included to provide a more complete understanding of the current state of the field. The result is a synthesis and summary of the open research challenges, as well as the opportunities, for mapping for autonomous driving.

Structure of the Review

This review is structured into six main sections: 1) general definitions; 2) uses of maps for autonomous driving; 3) data requirements; 4) production of maps; 5) privacy, security and standardization, and; 6) open research challenges (as illustrated in Figure 1).

Approach of the Review

The aim of this review is to provide a high-level discussion of the many facets of mapping for autonomous driving and to highlight the open research challenges in the field. The review begins with an overview of the uses of maps for autonomous driving and the different types of information that are used. Next, data requirements are explored, focusing on the quality, representation, format, and storage of the data. Following that, the article explores the production of the mapping data, specifically the hardware platforms and algorithms as well as issues of maintenance and update. Privacy, security, and standardization issues are then discussed in the fourth section. The review concludes with 11 open research challenges for mapping for autonomous driving.

As the review is intended to provide an overview of all aspects of mapping, it is not possible to cover every topic to exhaustion. For further details on each of the related areas, the following papers can serve as additional references: localization techniques [40], simultaneous localization and mapping (SLAM) [10], perception systems [13], coordinate systems and global navigation satellite systems (GNSS) positioning [22], georeferencing [24], and national-scale mapping [70].

General Definitions

What Is an Autonomous Vehicle?

As a generic definition, autonomous vehicles can sense the surrounding environment and navigate without human

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input. Under this wide category, there are many modes of autonomous vehicle, from air-based [44] to underwater-based [52] to underground-based [45]. This survey focuses on aboveground-based autonomous motor vehicles, which transport goods or people over land. These vehicles are also commonly known as *self-driving* or *driverless cars*.

> Maps can provide an important trusted baseline where the availability of sensors cannot be guaranteed.

What Is a Map?

There are many different definitions of *map*. Broadly, a map is a representation of real-world features and its location as objects. It is a symbolic and abstract depiction of both physical features (e.g., buildings, roads, and lakes) and nonphysical concepts (e.g., air pressure or happiness). For this review, an all-inclusive definition of *map* is adopted: a map provides information about real-world features or phenomena and its location within a 2D or 3D space. The terms *map* and *mapping* are used interchangeably as both verbs and nouns to describe the creation process and the end product—both aspects are discussed in this review.

For autonomous vehicles, maps can be created beforehand (often referred to as prebuilt, a priori, or offline maps) or in real time (online or simultaneous mapping). Prebuilt maps can contain much more detail and information than "conventional" maps created for humans, and, as such, are often referred to as "high-definition" (HD) maps. In this review, the focus will be on prebuilt and offline HD maps and the open research challenges surrounding the uses and production of such maps for autonomous driving.

Uses of Maps for Autonomous Driving

Information from maps supports many functions of autonomous driving, including self-localization, vehicle control, motion planning, perception, and system management. Maps for autonomous vehicles can provide static and dynamic information, such as road grade, curvature, and speed limits as well as traffic speed, traffic congestion, and temporary road works [87].

One of the leading uses of maps is self-localization—the process of identifying where your vehicle is in relation to the surrounding environment. By using range- or camera-based sensors, it is possible to localize yourself on the prebuilt map, using lane markings (Bao et al., 2016), landmarks (Sons et al., 2017), pole-like objects (Spangenberg, Goehring and Rojas, 2016), lidar intensity (Barsan et al., 2018), or the overall geometry of the surrounding environment [88], [89]. It has been suggested that even the underlying geology coupled with a ground-penetrating radar can be exploited for the purposes of localization [90]. Maps can also help predict the blocking, reflection, and diffraction of satellite signal availability for GNSS positioning [91], [92]. Once the ego-position is obtained, the prebuilt map can provide information that enables the system to make vehicle control decisions, such as lane change maneuvers, some of which are not possible with only sensor-based methods [2]. Furthermore, maps can provide additional robustness and foresight for motion planning, allowing the vehicular system to "see" ahead of what is within the sensor range. It can also help with distinguishing dynamic objects and detecting obstacles [25]. There are three main categories of mapping information that an autonomous driving system uses: topological, geometric, and semantic (as illustrated in Figure 2).

Topological Information

Topological maps can provide information on the connectivity between geometry features. In the case of autonomous driving, this is usually the network of roads. Existing navigation maps can provide this information, although new deep-learning methods can be also used to extract topology from other sources, such as aerial imagery [93]. At the macroscale, topological information can allow vehicles to traverse the most energy-efficient route, based on road grade, traffic speed, and intersection delays [87]. At the meso- and microscale, topology can ensure that autonomous vehicles obey traffic regulation orders, such as one-way streets.

Geometric Information

Beyond the topology of roads, it is important to consider the geometry or shape of other environmental features. Accurately representing the geometry of the objects and features around the vehicle is critical to many functions of autonomous driving. These features can be split into three categories: static, temporary, and dynamic features. Static features are physical objects that are immovable and, thus, represent a permanent obstruction. Examples include buildings, carriageways, curbs, pavements, traffic signs, traffic lights, lanes, junctions, bridges, and tunnels. Knowing where these static features are is not only useful for obstacle avoidance but also can provide reference points of localization within the urban environment [94]. Temporary features are physical objects that may exist at its location for only a limited amount of time. These include temporary road works, traffic cones, and parked vehicles. Notably, these temporary features are unlikely to be mapped, but their existence can mislead vehicle sensors or interfere with vehicle-to-vehicle (V2V) and

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vehicle-to-infrastructure communications. Dynamic features are moving vehicles, people, or objects. Again, while these may not be mapped, knowledge of their location and trajectory is critical for successful motion planning. Accurately mapping the static features can enable easier detection of temporary and dynamic features.

Semantic Information

Finally, alongside topology and geometry, semantic information can provide the "meaning" of features. Semantic information for autonomous driving includes road speed limit, lane information, and road classification. It can also provide relational information, such as how lanes work together, where vehicles can and cannot turn, and where vehicles must stop. By providing rich contextual information, autonomous driving systems can make informed decisions in different driving scenarios. For example, at a traffic light intersection, the on-board camera detects that there are two sets of lights. The traffic light to the left is showing a left green filter arrow, while the set on the

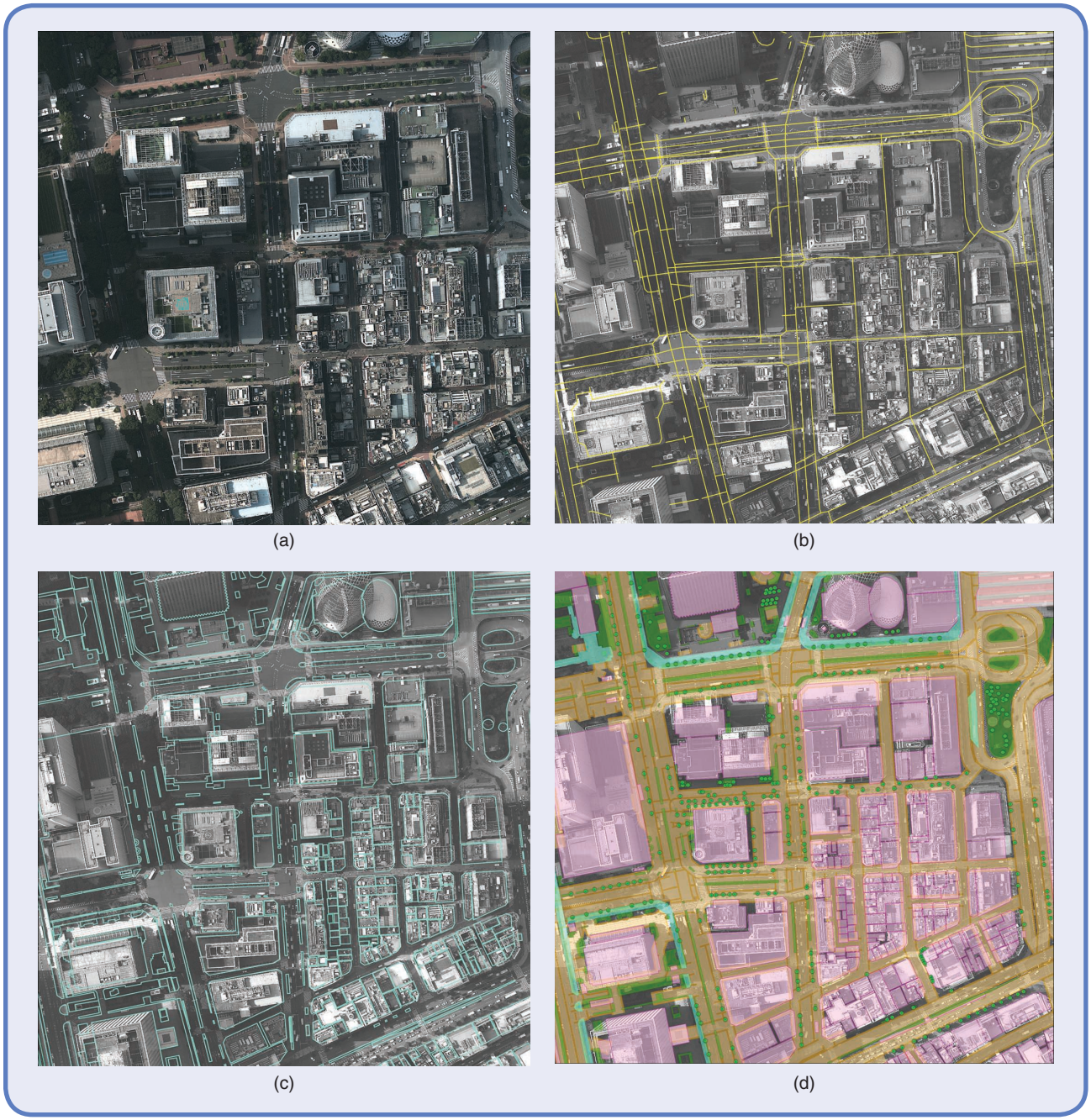

FIG 2 Examples of mapping information. (a) Satellite imagery, (b) topology, (c) geometry, and (d) semantic information.

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right is displaying a red stoplight. By linking the camera information with the map, the system can determine that the left set of traffic lights refers to the current lane the vehicle is in and, therefore, that it is safe to proceed in turning left.

One consideration is that semantic information must be attached to geometry. As a result, the underlying geometric data must be structured accordingly to accept the semantic information, e.g., if a single stretch of road has multiple speed limits, then the geometry must be split into separate segments. Therefore, for semantic information, a single 3D mesh is nonviable, and structured geometry (where features are clearly delineated) is preferred. Semantic information presents the pinnacle of autonomous driving maps. While it is information rich, the data are nuanced, and it is not straightforward to collect or incorporate. Recent advances in computer vision techniques have facilitated automatic semantic mapping approaches [68], [83].

Data Requirement for Autonomous Driving

Eliciting and formalizing data requirements is critical in designing data that are fit for purpose. As it is impossible to capture every single detail of the environment around us, map producers must select features and phenomena that are most important and useful for specific applications. Where this differs from traditional user requirement gathering is that the end user is a machine. Maps for machines differ from maps designed for humans, as they can contain much more information critical for decision making. Conversely, human maps are often simplified into abstract representations to enable easier and faster visual comprehension of spatial relationships. This begs the question: How can we design maps for a machine?

Many researchers have attempted to define the requirements of maps for autonomous driving. Gwon et al. [23] simplify the requirements into three criteria: centimeter-level accuracy, storage efficiency, and usability. The authors stress the importance of lane-level information, 3D geometry of the road, and ease of download and update of data via wireless networks. Similarly, Seif and Hu [67] recommend that maps have an accuracy of ±10 cm, enhanced by updates on the real-time status of the wider car environment (to a range of about 1 km). However, it is unclear where the specification is derived from, beyond an estimate.

It is important to note here that even within the category of autonomous driving, there will be differences in requirements depending on the scenario. For example, vehicles operating in a geofenced and controlled environment (e.g., autonomous highway or airport) will have different needs than a vehicle driving in a complex urban environment alongside human-controlled cars. Regardless of the context, it is expected that there will be some commonality across the different scenarios, e.g., drivable area [85].

Data Quality

Broadly, data quality is the degree to which data can satisfy a certain objective. Data quality is a pillar of autonomous vehicle maps, as reliable data are not only useful for the user but also critical to safety. Data quality in the context of autonomous driving can be posed as the question: "Do the data increase or decrease the risk of human life, damage to the vehicle or the third party vehicle, or property damage?" [85].

There are five key components to data quality for autonomous driving maps: 1) accuracy, 2) precision, 3) completeness, 4) consistency, and 5) currency. Accuracy is the discrepancy or deviance between the real-world value against the mapped value. Simply stated, how close is the observation to reality? Autonomous driving maps tend to focus on positioning accuracy, which can be absolute or relative. Absolute accuracy describes how the objects and features are positioned on the map with respect to their true positions on the ground with reference to a coordinate system, e.g., WGS1984. In contrast, relative accuracy is a measure of how objects are positioned in relation to each other, e.g., two lampposts are within ±15-cm tolerance of each other. Relative accuracy tends to be higher than absolute accuracy. For autonomous driving, meter-level absolute accuracy could identify which street you are on, whereas centimeter-level accuracy could detect which lane you are in. Within the current commercial offerings, there is no consensus on the minimum accuracy requirements for autonomous driving maps. Table 1 displays a summary of the stated absolute and relative accuracies of different commercial autonomous driving maps currently available. Note that this table is not exhaustive, as some organizations do not publish the full specifications of their products.

The second component of data quality is precision (also known as *resolution* or *granularity*). *Precision* refers to how exact the description of the data is and is the smallest discernible unit or smallest unit represented. For example, a precision of 5 cm means that objects and details down to 5 cm in size will be on the map, but anything smaller will

Table 1. An example of absolute and relative accuracies of commercial maps for autonomous driving.

Map	Absolute Accuracy	Relative Accuracy	Source
AND	15 cm	5 cm	[3]
Civil Maps	15–20 cm	1–5 cm	[14]
HERE	10–20 cm	—	[65]
lvl5	5–50 cm	—	[46]
Sanborn	7–10 cm	—	[63]
TomTom HD Map	1 m or better	15 cm	[74]

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not be distinguishable. It is important to note that precise data can be inaccurate, and that accuracy and precision are not synonymous. In the context of autonomous driving, more precise maps do not always equate to better maps. Since no measurement system is infinitely precise, all maps have a limited resolution [77]. Furthermore, raw sensor data may be generalized through simplification of aggregation to reduce the complexity and size. This, in turn, reduces the computation load and storage requirements, but certain useful details necessary for autonomous driving functions could be lost. In other cases, however, precision may be a hindrance. For example, a point cloud map may be up-sampled to remove noise and outliers or to reduce the effects of local minima in map matching.

The third component of data quality is completeness. This refers to the degree to which the map contains all of the real-world features. In other words, are all possible objects represented on the map? Completeness is often measured by omission (data absent) and commission (excess data present). Analysis based on incomplete data will lead to erroneous results and decision making. For example, if a tall building had been recently erected but not reflected in the mapping data, then a system relying on 3D map-aided GNSS positioning may not be able to correctly predict the blocking and reflecting of satellite signals [17].

The fourth component is consistency. This is the degree of adherence to logical rules of data structure, attribution, and relationships [31]. For example, within a topological map, do all roads connect at nodes? Do all flows point in the correct direction? In the case where the road directions are not consistent in the map, this lack of consistency could lead to traffic violations committed by the autonomous vehicle or, worse, a collision with another road user.

The final component is currency (also known as *temporal accuracy*, *up-to-dateness*, or *freshness*). This refers to the time between the creation of the map and the use of it. Prebuilt maps are outdated as soon as they are created. Even with the most accurate, precise, complete, and consistent map, if it is out of date, it can become unreliable. The extent to which currency affects the reliability of the map is application specific. For example, long-term applications, such as urban planning, could use data that are two to three years out of date without impacting the resulting analysis. For the safety-critical application of autonomous driving, however, data that are one or two days old may not reflect the latest obstacles in the road environment. For example, a new set of roadworks may have appeared overnight, or a temporary roadblock may have been put in place. To counter this, one option is to include a temporal measure in the metadata, whereby the older the map, the less reliant the autonomous driving system is on the data. Another option is to have constant dynamic and incremental map updates. When a vehicle discovers a change in the environment, the sensors map the new geometry and propagate the updated geometry, semantic, and topological information to other vehicles through the connected environment. The issue of map update and maintenance will be discussed in a later section of this article.

Measures of Data Quality

Mapping is often produced by many different organizations, in both the public and private sectors. The following question arises: How can we discern which map is "better?" Data for mapping can be created using different methods, processed using different techniques, and produced at different times of year. These factors can lead to discrepancies in quality between the resulting maps.

As there is yet to be a consensus on the final format(s) of autonomous vehicle maps, there has been little research on measuring the data quality of mapping as well as the subsequent impact of quality on the functions of autonomous vehicles. Adjrad et al. [1] investigate the impact of 3D data quality on GNSS positioning. The authors conclude that using 3D city models with complete geometric detail gives significantly higher accuracy than simple block models, and missing buildings lead to larger positioning errors. Wong and Ellul [79] proposed the use of geometry-based metrics as part of fitness-for-purpose evaluation of 3D maps. Javanmardi et al. [34] define four criteria to evaluate a map's capability for vehicle localization: feature sufficiency, layout, local similarity, and representation quality. In a follow-up study, Wong et al. [80] use 2D geographic information in lieu of 3D point clouds to evaluate the environment. By assessing the impact of the map, it is possible to identify areas where self-localization may fail, allowing a human driver to take control.

Dealing With Errors in Mapping

As autonomous vehicles become increasingly reliant on maps, what happens when the map is wrong? Maps will never be entirely free of errors. At what point is the quality of a map "good enough" for its purpose?

In the previous section, it was suggested that the map could provide a reliable backup to other sensors. However, a poor-quality map may provide misleading information. This could lead to incorrect vehicle trajectories or misguided decision making, compromising the overall safety. Murphy and Pao [53] propose an approach to detecting unmapped or incorrectly mapped roads and parking lots in the context of map matching. The result is a system that is more robust to map errors and can even propose corrections to errors in the underlying maps. From a similar, but alternative, angle, Yang and Huang [83] investigated how to make autonomous vehicle systems more resilient to malicious attacks on its sensors. Ultimately, it is important that there is adequate redundancy—if a map-reliant vehicle function fails due to invalid data, another nonmap sensor

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must serve as a fallback until a human driver can retake control.

> From the data producer perspective, a single map would result in lower overall costs in production, maintenance, storage, and update.

Representation and Data Format

There are endless ways to model and represent the same real-world phenomena at different scales, with different levels of accuracy, and at different levels of detail. A road, for example, can be modeled simply by its centerline as a linear representation. This is an efficient way to model its connectivity and topology. Alternatively, it can be modeled by an aerial representation, with precise lane-level information, lane markings, and road gradient and curvature. More complex mapping frameworks may use a combination of representations. For example, "Lanelets" use a mix of points to represent vertical structures, polylines to represent the bounding lines of the drivable area, and areas to show sections of the map in which undirected or no movement is possible [7], [58]. The choice of representation highly depends on the objective of the data—in other words, the data must be modeled in a way that is fit for purpose or fit for use.

Huang et al. [29] loosely define autonomous vehicle maps into two categories. The first group is strongly related to the raw sensor data, such as 3D point clouds. While these maps are computationally efficient and consistent for specific sensors, they are often not recognized or cannot be used by other sensors. Furthermore, the raw data are only stable for a short time with similar environmental conditions. The second group of maps represents the environment in a more generalized form. This presents a more interoperable mapping solution at the cost of a slight reduction in efficiency across all sensors.

One question that remains is whether a traditional "one-size-fits-all" approach to mapping is appropriate or if multiple maps tailored to specific functions of autonomous driving is a better approach. From the data producer perspective, a single map would result in lower overall costs in production, maintenance, storage, and update. While multiple maps may lead to increased cost, this can be ameliorated if they can be derived from a single master source of data. This also ensures consistency between the different layers. Producing multiple maps has the advantage of being able to tailor the level of detail and information for different functions of the autonomous vehicle. For example, HERE [26] uses three layers within its live HD map: 1) Road Model, 2) HD Lane Model, and 3) HD Localization Model. The Road Model contains the road topology, centerline geometry, and road-level attributes. This is useful for macroscale route planning and navigation. The HD Lane Model contains lane-level topology, geometry, and attributes, allowing finer-scale motion control and path planning. Finally, the HD Localization Model includes object-level data and generic localization data to assist the vehicle in accurately locating itself within the lane in which it is traveling. One disadvantage of multiple maps is harmonizing updates and ensuring topological, geometry, semantic, and temporal correspondence.

Alongside the choice of representation is the selection of data format. A recent report conducted by Zenzic [85] in collaboration with Ordnance Survey (the national mapping agency of Great Britain) explored more than 30 different formats of mapping data for autonomous driving. Four formats were recommended: 1) LAS 1.2/LAZ for point cloud data; 2) OBJ for terrain and 3D objects, such as buildings; 3) OpenDrive for track-based road networks; and 4) ESRI shapefiles for representing a wide range of specific key features and their attributes. Further work is required to validate whether these formats can satisfy all of the requirements for autonomous driving.

Storage of Maps

Maps for autonomous vehicles contain far more detail than traditional maps, such as lane dimensions, distance from pedestrianized areas, and curb height. These autonomous vehicle maps require enormous computational and storage power to create and handle in real time [30]. For example, a 3D point cloud for a 300- × 300-m area may consist of more than 250 million points, with a single vehicle estimated to produce anywhere from 11 to 152 TB of data per day [61]. For reference, the Apollo project from Baidu currently uses a modified industrial PC configured with a single 1-TB hard disk drive [9]. Maps for autonomous driving may also contain multiple interconnected layers, e.g., one layer for localization and another with information on lane geometry and connectivity, further compounding the size of the data. In addition to the vast data sizes, maps for autonomous vehicles come in a wide range of representations, from discrete vectors to 3D point clouds to voxels. Where multiple maps are used simultaneously in a single system, there must also be adequate measures in place for ensuring correct correspondence between the data layers. These factors create challenges in how to store, maintain, and update such vast and heterogeneous data sets.

Map storage systems for autonomous vehicles must be able to: 1) handle massive volumes of incoming and outgoing data and 2) handle multiple formats simultaneously.

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By dynamically streaming and downloading mapping content, the issues relating to local storage and updates are partially resolved.

Although traditional data storage methods can support this to certain extents, they are currently unable to meet all of the requirements of autonomous driving. The development of solid-state drives (SSDs) has provided a low-latency and high-capacity data storage option. In 2018, SSDs were available in sizes up to 100 TB [55], but 120–512 GB were more common. As autonomous vehicle maps grow from local to regional to national scale, the storage required to hold the mapping data will only increase. One approach is for data reduction through compression. Dymczyk et al. [16] present a map compression algorithm, reducing the map to only landmarks that are deemed most useful for place recognition. This not only improved the scalability of the system but also reduced the computational load.

Even if the on-board storage is large enough to hold all of the necessary mapping data, the data itself must be fast to access. For this, the structure of the data itself is important to consider, whereby the logical organization of the information can facilitate its use while preserving its integrity. In other words, it is imperative that the data are organized in such a way that it is fast to look up and retrieve. One option is to employ a spatial index (e.g., Quadtree, Octree, *R*-tree, *k*-*d* tree). This arranges the data such that objects close to each other in the real world are also closer within the storage medium. Therefore, in a tiled mapping architecture, tiles that are adjacent to your current location would be the quickest to load. Without spatial indexing, the system would require a sequential scan of every record within the whole database, resulting in a much longer processing time. The disadvantage of using indexed data is decreased performance in updates and changes within the system, as the indices themselves also require an updating. This may be an issue for autonomous driving maps, which require dynamic, continuous, and real-time updates.

Increasingly, the trend is moving away from locally stored maps to cloud-based maps as a service. By dynamically streaming and downloading mapping content, the issues relating to local storage and updates are partially resolved. Maps no longer have to be fully stored using on-board storage, allowing vehicles to obtain only the necessary mapping information on the fly. At the same time, only the central source of mapping data requires updating, avoiding data conflicts and ensuring consistency between the mapping data in different vehicles. A cloud-based map service, however, brings different challenges in the delivery and dissemination of the data. Issues include latency, loss of connectivity, and uploading locally detected changes back to the central map server. Furthermore, the geographic distribution of vehicles can also be problematic. To serve vehicles on a regional or national basis, there would need to be multiple servers holding duplicates of the same data to ensure minimal latency. V2V and infrastructure-to-vehicle communications could assist in content delivery in a cooperative environment, whereby the most popular content is cached temporarily and shared among all users.

Production of Autonomous Driving Maps

Mapping Hardware Platforms

Data for creating maps can be captured in many ways, from satellite to aerial to terrestrial sources. These sources may capture a wide variety of data, from high-resolution red, green, blue imagery to 3D point clouds to thermal imagery.

For autonomous driving, the focus is on operating in the road environment. As such, mobile mapping systems (MMSs) are often a popular choice, as they map at the road-level perspective. MMS are specialized vehicles equipped with high-end sensors (e.g., lidar scanners and stereo cameras) and positioning sensors (GNSS and inertial measurement unit/inertial navigation system), which can generate highly detailed 3D point clouds of features, such as building facades, roadside infrastructure, and trees. Simultaneously, high-resolution images can be captured using a single or multiple cameras and using wide-angle or fisheye lenses to cover the full 360° field of view [25]. These sensors can be used together to capture dynamic environments, traffic-related information, traffic signs, and road surface conditions [76].

In recent years, however, there has been a shift away from using a single specialized vehicle equipped with expensive sensors to deploying many and multiple vehicles with cheaper and lower-end sensors [20], [37]. Through this cooperative approach, multiple vehicles working simultaneously as a team can produce many local maps, which are then merged into a single global map [86]. This has the advantage of not only being lower in cost but also providing more flexibility in data collection. The overall system is more robust, since if one mapping vehicle fails, the others can still operate. Furthermore, this can reduce the accumulated error often suffered by single-vehicle mapping algorithms. This approach can also be extended to crowdsourcing, whereby multiple sources of low-fidelity data can be compensated through aggregation [37]. For example, Ruhhammer et al. [62] use crowd trajectory data to extract information about

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intersections and traffic lights. The multivehicle, crowd-sourced approach is much more scalable and has been adopted by several companies, including HERE and lvl5 [9], [46].

Aerial-based mapping platforms can provide a complementary approach to MMS.

Aerial-based mapping platforms can provide a complementary approach to MMS. While MMS excel at capturing road-level assets, aerial-based techniques can collect accurate information on features not visible from the road, such as roofs and areas behind buildings. In addition, aerial data have an overall better global accuracy and are less sensitive to errors due to signal loss. Unmanned aerial vehicles (UAVs) can be used as a low-cost and flexible data acquisition technique for extracting key information, such as road maps [82], lane markings [19], and landmarks [29].

Dynamic Objects During Data Collection

One persistent issue for data collection of autonomous driving maps is the presence of dynamic objects, e.g., other vehicles and road users. These dynamic objects can block the sensors, resulting in "gaps" in the data, making the data noisy and unreliable. While dynamic objects may not necessarily need to be mapped, they need to be identified and removed. Feng et al. [18] propose the use of cameras to detect vehicles and then retrospectively remove them from the point cloud automatically. Similarly, Romanoni, Fiorenti, and Matteucci [59] use a single lidar and camera images to generate a textured 3D mesh, whereby moving objects do not appear in the final results.

Algorithms for Mapping

As discussed in the previous section, there are, broadly, three types of information used for autonomous driving: 1) topology, 2) geometry, and 3) semantic. As topological information is commonly available in many open and commercial offerings, the current focus is on the production of accurate geometry and detailed semantic maps for autonomous driving.

Geometric information expresses the shape and position of the object on the map. Regardless of the format of the data (e.g., point cloud or structured polygons), if the positions of all data elements that represent the object are accurate, then the geometric information of the object will also be accurate. As current MMSs can generate highly detailed 3D point clouds and capture high-resolution images, the challenge is how to localize these representations accurately.

One solution is SLAM, which can build a map in an unknown environment at the same time as keeping track of an agent's location within the created map [5], [15], [56]. SLAM has made significant progress in the last two decades and become a key component in robotic navigation. The first and second milestones of SLAM are the extended Kalman filter-based SLAM (EKF-SLAM) [69] and particle filter-based SLAM [51]. EKF-SLAM and particle filter-based SLAM are more popular due to online localization.

In recent years, graph-based SLAM methods have undergone a renaissance and currently belong to the state-of-the-art techniques with respect to speed and accuracy in mapping [73]. For wide-area mapping problems, it was found that GraphSLAM can handle a large number of features and even incorporate GPS information into the mapping process [73]. Moreover, Levinson and Thrun [43] integrated GPS, IMU, wheel odometry, and lidar data to generate high-resolution 2D road surface maps using the GraphSLAM method. The integration of GPS information is particularly important in ascertaining the global position. GPS information, however, can be problematic in urban areas, where the accuracy of GNSS degrades significantly. One alternative is to use ground control points to manually calibrate the MMS data. Another approach is to automatically identify the correspondences between the local and global data sources, such as using road markings [33], [38].

The algorithms of extracting semantic information fall into the scope of object detection and recognition from images and point cloud data. There are many efforts on extracting road markings from aerial images to build the road model [4], [28], [39]. Other roadside assets, however, are more difficult to extract. Vertical objects, such as traffic signs and signals, are difficult to "see" from above in nadir aerial imagery and can be too "thin" to capture fully by an MMS. With developments in computer vision and deep learning, the aim is to import the semantic information into a map using the following steps: 1) identifying the relevant objects and recognizing the scene semantics automatically from images captured from MMS and 2) extracting the corresponding objects from the 3D point cloud map [47]. There are a number of computer vision approaches to achieve this, including convolutional neural network for traffic sign detection [42], fully convolutional network for lane segmentation [32], and PSPNet for scene segmentation [72].

While deep learning has dramatically improved in detection accuracy over recent years, it is still not possible to deliver 100% accuracy on semantic information extraction. It is, therefore, important to consider the

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It is vital that any changes in the real environment are reflected in the map as soon as possible to avoid vehicular errors derived from inaccurate mapping.

reliability of the results and its possible impact on autonomous driving, e.g., if a speed limit sign is misclassified, what are the ramifications?

Maintenance and Update

Maintenance and update are critical considerations for any data system. The update process may need to handle data coming from a variety of sources and at different scales. It will also need to recognize equivalences and differences between the new data and stored representations. It must be able to capture any new changes in the environment, e.g., temporary road works, and propagate it through all data layers. The ever-changing environment and road structures mean that keeping maps updated remains a challenge [60]. For example, Pauls et al. [57] reviewed 80 km of German highways, showing that 41% contained outdated information. Changes included new lane markings, guardrails, and whole road surfaces as well as complete reconstructions.

Near-real-time crowdsourcing of update information potentially could provide the solution, e.g., Mobileye's Road Experience Management system [50]. Similarly, Mapper employs a network of freelancers (who may also drive for ride-sharing services) to map different areas on request [48]. By exploiting a distributed mapping workforce, maps can be kept up to date, and additional information on long-term but temporary features (such as roadblocks) can also be incorporated [85]. Unlike traditional map making, the maintenance and update of autonomous driving maps will be a continuous and ongoing process. It is vital that any changes in the real environment are reflected in the map as soon as possible to avoid vehicular errors derived from inaccurate mapping.

Exploiting Other Sources of Spatial Information

As maps have been produced for and used by humans for centuries, are there existing sources of information that we can exploit to produce maps for autonomous vehicles? Conventional or "traditional" maps are often described as inaccurate and lack the detailed geometry required for localization and navigation. However, it would be remiss to disregard this rich (albeit more generalized) source of mapping data. In both the private and public sectors, there is a wealth of information available that can be used to enhance the production of maps for autonomous driving. For example, geographic information systems (GIS) can provide a useful source of spatial information at the local, regional, and national scale.

Data from national mapping agencies or OpenStreetMap can provide invaluable information on road layout, road topology, roadside features, and building footprints. In addition, GIS data sets can provide rich semantic information not available from sensor data. Despite being produced at a coarser level of accuracy, these data sets can be integrated with other sensor information to generate accurate lane-level maps [36], supervise road detection [41], define driving corridors [21], and estimate vehicle dynamic states [35]. Advances in deep-learning approaches have also enabled existing street-view image databases and online navigation maps to be exploited for road layout inference [66]. The use of "conventional" maps, however, is not without its problems. These data sets can be inaccessible and of varying quality, making them difficult to use and potentially unreliable without thorough quality control.

Privacy, Security, and Standardization

Privacy and Security

The production of increasingly comprehensive and accurate maps raises many issues of privacy. The data that will be generated from autonomous vehicles will be incredibly detailed and vast. How can we ensure that the users' data are protected and their privacy is preserved? While detailed mapping data bring many benefits, concerns also arise on who owns this information, who controls it, and how it is used [8]. It also raises important questions of competition regulation and public safety [11]. Schoonmaker [64] highlights the current inadequacies of protecting location-based data on users in a connected vehicle environment. However, Taeihagh and Lim [71] argue that overly stringent and excessive privacy regulations may impede technological advancement. Ultimately, data must be unlocked to generate services for users and to enable safer and more reliable autonomous driving. Boeglin [8] argues that autonomous vehicles should infringe on user freedom and privacy to the same extent as the reductions in liability for the users of self-driving cars. Furthermore, any social costs incurred by forfeiting these values must be outweighed by administrative efficiencies or other identifiable social benefits.

Alongside privacy, security is another critical consideration for the successful deployment of autonomous vehicles. Autonomous vehicles are reliant on a lot of hardware and software to ensure passenger safety—to that end, intelligent transport systems must be immune to harmful threats imposed by humans through the introduction of viruses, remote intervention over wireless networks, or even sensor interference. Researchers have shown that attacks on passenger

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vehicles are possible and can have severe consequences. Miller and Valasek [49] demonstrated that it was possible to remotely take complete control over the functionality of an unaltered 2014 Jeep Cherokee. This resulted in Fiat Chrysler recalling 1.4 million vehicles [6]. Wouters et al. [81] showed that standard radio equipment could be used to break the encryption on a Tesla Model S's keyless entry system by wirelessly cloning the key fob. In response, Tesla had to create a new iteration of its key fob to patch this underlying flaw.

Defining a Common Data Standard

Standards are important, as they provide people and organizations with a mutual understanding, facilitating communication, interaction, and compliance with laws and regulations. It enhances interoperability of products from different manufacturers and can speed up the introduction of new technologies to market. Once there is a strong understanding of requirements, the next step is to develop and establish a set of data standards.

Currently, automobile manufacturers, map producers, and research institutions are individually creating their own map formats tailored specifically to their automated driving system [2]. This is problematic, as not only is there a duplication of effort in the mapping task but also data from other sources may not be interoperable (e.g., a map created by Toyota will not be compatible with a BMW vehicle). Having individual and custom formats is inefficient and results in a high cost per usage of the data. One potential solution is to develop a national standard for mapping for autonomous driving—but who should be responsible for setting the standard?

A recent report by the British Standards Institution identified a total of 661 standards published by countries and standard-developing organizations relating to different aspects of connected and autonomous vehicles. The report found that while existing standards (such as *ISO 26262–Road vehicles–Functional Safety*) were already widely used in the automotive industry, they required significant adaptation and revision to accommodate for the needs of connected and autonomous vehicles. Notably, none of the standards specifically defines the requirements for mapping. The breakdown of the review is presented in Table 2.

In April 2018, multiple Chinese governmental ministries jointly issued the national "Administrative Rules on Intelligent and Connected Vehicle Road Testing" for China. This was in response to the fact that local standards varied from region to region, posing barriers for companies looking to conduct autonomous vehicle tests across the country. However, within these guidelines, there was no mention of specific map requirements.

In the United Kingdom, the British Standards Institution, Transport Systems Catapult, and the Centre for Connected and Autonomous Vehicles are actively working with industry and government on developing a set of standards [85]. As part of this, Ordnance Survey, the national mapping agency of Great Britain, is looking to contribute to the geospatial and mapping aspect of the objective.

In Japan, the Cross-Ministerial Strategic Innovation Promotion Program has been testing the specification and accuracy required for HD maps. Between 2017 and 2018, autonomous driving maps were created for 14,000 km of expressways. As part of an effort of standardization, Japan has participated in industrial initiatives, such as Open AutoDrive Forum and Navigation Data Standard [78].

The development of a national or international standard could enable the establishment of a common map environment. In this scenario, government agencies, private organizations, and automobile manufacturers could work together to provide a single source of authoritative and neutrally hosted mapping information. This, in turn, would increase interoperability and confidence in the source of the data. By providing mapping as a common service, economic efficiencies can be achieved (in what is already an expensive endeavor).

Open Research Challenges

As a result of the aforementioned survey, 11 open research challenges were identified during the review and are presented below.

Challenge 1—What Should We Map?

The first, and most important, challenge is to identify what information is critical to include within a map (or maps) for autonomous driving, regardless of the representation or method of production. This can be reformulated as the following questions. What are the different types of data that need to be stored in the map? What are critical "must haves" and what is additional "nice-to-have" information? Of course, the data requirements for autonomous driving maps will differ depending on many factors, including the vehicle platform, environment, manufacturer, and country of operation. However, there will be commonalities across different maps for basic information (such as road geometry). The identification of the core information that can be used by many different vehicular systems and platforms should form the first step toward defining clear data requirements for autonomous driving.

Table 2. A count of international standards for connected and autonomous vehicles, by topic.

Topic	Count
Connectivity/connected vehicles–technology	244
Awareness	123
Connectivity/connected vehicles–applications	91
Localization standards	84

Note: adapted from [12].

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Challenge 2—Mapping Traffic Laws and Regulations

A key challenge for mapping is the inclusion of local and national traffic laws and regulations in the map. While autonomous vehicles equipped with sensors can detect traffic signage and road markings, if the sign is missing or the road marking is obscured, the vehicle is still under obligation to obey the rules of the road. To that end, it is crucial to supply autonomous vehicles with a consistent source of information for traffic regulations.

For autonomous driving, there is a decision-making challenge. For example, how strictly should the rules be followed, e.g., should the vehicle cross a set of double solid white lines to overtake a broken-down vehicle? From a mapping perspective, as autonomous vehicles are accountable for traffic laws at local, regional, and national levels, the heterogeneity of rules and regulations (as well as data quality) makes it difficult to create a consistent and singular set of data. Furthermore, the representation poses issues too, for example, where should the boundary of a speed restriction begin? The adherence to traffic laws is also often culture dependent and is particularly notable in less developed countries. For example, understanding the underlying road structure to operate an autonomous vehicle in India is very different from operating one in the U.S. This is a major challenge for generating map structures that are valid internationally. The challenge of mapping traffic laws and regulations must be considered fully to enable to successful operationalization of fully autonomous vehicles at the local, national, and international scale.

Challenge 3—Improvement of Navigation Information Integrity Through Redundancy

Autonomous vehicles have many sensors to enable them to navigate the environment safely. The multiple sensors within the perception system mean that a certain amount of redundancy is built in to account for sensor failure. In some cases, it is possible to combine the data through sensor fusion, thereby enhancing reliability.

If the sensors present conflicting information, however, this poses a certain issue: Which sensor should we trust more? Any mapping data used by autonomous vehicles must be reliable and authoritative, so that decisions made by the vehicle control system comply with national and local laws [85]. For localization, not only is accuracy important but also integrity. Integrity provides a measure of trust that can be placed on the "correctness" of the information. It focuses on ensuring that erroneous values are not dangerously far from the actual values. In the case where there is conflicting sensor information, if the "wrong" sensor is relied on, it could cause a driving error. Could mapping information provide the necessary tiebreaker to resolve these conflicts?

Challenge 4—Intelligent Driving Style

Within different road environments, autonomous vehicles may need to adapt their driving style. For example, in a highly urbanized environment, there may be a lot of stop-and-start city traffic with a higher number of vulnerable road users. There may also be many tall buildings with mirrored glass facades that interfere with GNSS sensors. Conversely, rural environments may be characterized by curvy and mountainous roads, with blind corners. Rural areas may also lack enough features for effective map-based localization. The question, therefore, is the following: Is it possible to derive information from the map for the autonomous vehicle system to intelligently adapt the driving style? A simple example would be for the system to recognize from the map that there is a steep downhill gradient ahead. In response to this, it engages and maintains a lower gear to take advantage of engine braking to slow down the vehicle during the descent. This, in turn, reduces the need to apply friction or magnetic brakes, lowering the risk of the brakes overheating.

From the mapping perspective, the challenge is twofold. First is identifying the specific relevant information to include within the maps to allow the system to anticipate the future driving environment, e.g., for the downhill braking example, the map will need the gradient information incorporated or the 3D road geometry to calculate it. The second challenge is for the map to provide the information in a timely manner, allowing the system to foresee any imminent change in the environment. This is an optimization challenge, as we want to allow the vehicle to "see" as much of the surrounding area as possible without inundating it with excessive amounts of data. This, in turn, is related to Challenge 8, on the storage and dissemination of maps.

Challenge 5—Defining Minimum Data Quality Requirements

Data quality is a critical consideration in the production of maps for autonomous driving. The challenge now is to ascertain the minimum data quality requirements for successful and safe autonomous driving as well as evaluating the impact of poor data quality on the functions of an autonomous vehicle. It is important to remember here that data quality is relative to the specific application, and that minimum data quality requirements for different autonomous driving functions may not be the same. Alongside minimum requirements, there is a need to develop suitable measures of data quality for autonomous driving maps as well as improving autonomous vehicle functions to be more robust against mapping errors.

Challenge 6—Defining a Universal Mapping Format

While it is generally acknowledged that maps will be a critical enabler of autonomous driving, what is not yet clear is the number of maps or mapping layers required, the representation, or which format works best for different functions. It is unlikely that one map would meet all of the complex requirements of autonomous driving, but at the same time, 50 different

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maps for 50 different functions would present huge challenges in computation, storage, maintenance, and update. Perhaps the challenge now is to identify the smallest number of mapping formats that can satisfy all of the necessary static and dynamic mapping information required for autonomous driving.

> The development of a common mapping standard will be the enabler of autonomous driving from the test environment to the real world.

Ultimately, the aim will be to standardize all mapping data related to autonomous driving. The development of a common mapping standard will be the enabler of autonomous driving from the test environment to the real world. As there is currently no universal mapping standard to cover all uses of autonomous driving, there is now an opportunity for organizations from industry and academia to come together and help bridge the gap between research and market. This task, however, is not easy, as standardization is an act of voluntary cooperation. Organizations must also avoid the pitfall of multiple and disparate efforts, as this can lead to numerous competing standards for the same technology, reducing the significance and impact of standardization.

Challenge 7—Collaborative Mapping

One proposed solution to mapping for autonomous driving is crowdsourcing or collaborative mapping. Commercial efforts by MobilEye [9] and TomTomxDENSO [74] are two examples whereby maps are created by leveraging intervehicle cooperation. The underlying concept is to use a group of individual vehicles to collect information about the environment in real time and transmit it back to a central location to be aggregated. As discussed previously, the main advantages include scalability, reduction of error accumulation, increased robustness, and decreased costs.

The main challenge for collaborative mapping, however, is the process of aggregation and reconciliation of the continuous stream of information. How can we integrate the many different sources of data into a single, trusted master data set? What is the best practice when reconciling data conflicts if two vehicles present slightly different information in the same time period? What if the two sets of information are wildly varying?

Since there are many vehicles involved in the mapping process, how should we deal with varying levels of fidelity and resolution? Should an accident occur due to a fault in the mapping, who is responsible? While collaborative mapping certainly has its strengths, there remain many challenges to resolve.

Challenge 8—Efficient, Effective, and Reliable Storage and Dissemination of Autonomous Driving Maps

The main challenge for the storage and dissemination of autonomous driving maps is how to deliver such voluminous data at speed without compromising on quality. Further research is required to investigate what is the most efficient and effective framework for storing and sharing mapping data for autonomous driving and how to build and maintain such infrastructure.

Challenge 9—Building Maps at the National or International Scale

For successful operationalization of autonomous vehicles, it is imperative to produce maps at city, region, and national levels. However, building a map that can scale well, be easily updated, and work under all environmental conditions is not a trivial task [10]. Research is now looking at producing maps with wider coverages [54], but many challenges remain in creating national or international maps. Part of the challenge is how to account for the local and regional differences in road geometry, signage, and road markings as well as traffic laws and regulations.

Challenge 10—Update and Maintenance

The update and maintenance of maps for autonomous driving will be a continuous and ongoing challenge. Part of the challenge will be to identify what triggers change in a map and at what level of change does an area require remapping. It is also important to identify the update frequency of maps for different functions and whether maps need to be updated monthly, weekly, daily, or even minute by minute. These considerations, in turn, raises issues of cost and the need to identify the most efficient update process without compromising on data quality and safety.

Challenge 11—Preserving Privacy in a Shared and Connected Environment

In the increasingly connected world, the challenges around privacy are ever prevalent. For autonomous driving, the requirement for centimeter-level positioning accuracy can infringe upon a user's locational privacy rights. As an individual travels through public space, there is a reasonable expectation that their location and movement will not be systematically recorded for other uses. For location-based services, however, the strength derives from knowing exactly where the individual is. For example, by collecting information destination, route, speed, and time traveled, the system can provide live updates on traffic congestion to all. Although revealing the ego-location can offer a wealth of opportunities, it can also expose the

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individual to targeted marketing by third parties as well as potential risks from less scrupulous citizens. The following questions then arise: What level of locational confidentiality should the user expect from autonomous driving, and what benefits should the user gain from any degradation in their privacy?

From a mapping perspective, can autonomous driving operate safely if the individual's location is to be obfuscated or entirely hidden? Should a map hold knowledge from previous journeys to enhance the functions of autonomous driving, or should user privacy be at the forefront? These are particularly important to consider within a collaborative mapping environment, as discussed in Challenge 7.

Conclusion

This article provided an overview of mapping for autonomous driving and presented eleven open research challenges. It has shown that while progress has been made, there remain many challenges and unknowns in the production, use, and regulation of autonomous vehicle maps.

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